

FINAL REPORT

Course: Applied Linear Algebra for IT

Course Code: 501032

I. REGULATIONS

This report is a replacement for the final written exam in the first semester of the academic year 2024-2025. Students need to adhere to the following regulations:

- Each report is conducted by a group of **one or two students**.
- For each question, students are required to present **detailed calculation steps**, and the final results.
- The report must be submitted in **PDF format**, and the content must be written based on the report/essay format of the Faculty of Information Technology. **In case students do not follow the Faculty's format, they will receive 0 points for the Report.**
- Each group of students must complete the **self-evaluation form** using the following template:

- <https://docs.google.com/spreadsheets/d/1WkFQLo6uvYalRSqT9dbsryC9QFjScKIV/edit?usp=sharing&ouid=104602653034506388459&rtpof=true&sd=true>
- Download the form as an Excel file (.xlsx extension)

- The report must include the following contents:

a. Chapter 1: Solutions (9.0 points)

Write **detailed calculation steps** and the final results of each question (in the section III). In each calculation step, insert the related Background knowledge (theorems, laws, rules, tests...) in case you apply it, for ex.:

- By Gaussian Elimination algorithm, ...

- Unusual cases (such as identical patterns, signs of plagiarism, etc.) will be scheduled for online discussions with the instructor.

II. Submission guideline

- Filenames of the report and self-evaluation files must be the **Student IDs**, for ex.,
 - A group of only one student with student ID 521H1495 will submit a self-evaluation file named **521H1495.xlsx** and a report file named **521H1495.pdf**
 - A group of two students with student IDs 521H1234 and 522H4321 will submit a self-evaluation file named **521H1234_522H4321.xlsx** and a report file named **521H1234_522H4321.pdf**
- Each student in a group submits **two files** to the "**FinalReport**" assignment on Elearning website of the Theory class.

III. Questions

Question 1 (4.5 points). Investigate in-detail the singular value decomposition (SVD) of a rectangular matrix:

- Definition/Formulas of singular values
- At least one proposition related to singular values
- Definition/Formulas of SVD
- At least one theorem related to a SVD finding
- Steps for finding a SVD
- Five applications of SVD
- Take two examples of SVD finding (examples from Internet/books are not allowed) :
 - o One with a matrix of size 2 rows $\times$ 3 columns and the rank of the matrix is 2
 - o One with a matrix of size 2 rows $\times$ 3 columns and the rank of the matrix is 1

Question 2 (1.5 points). Find a matrix P that diagonalizes $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Question 3 (1.0 point). Use the Gram-Schmidt orthonormalization process to transform the basis $S = \{u_1 = (4; -8; 8), u_2 = (8; 8; 4), u_3 = (8; -4; -8)\}$ for $\mathbb{R}^3$ into an orthonormal basis.

Question 4 (2.0 points). Consider the vector space $\mathbb{R}^3$ with two bases:

$$\varepsilon = \{ \varepsilon_1, \varepsilon_2, \varepsilon_3 \} \text{ in which } \varepsilon_1 = (1, 0, 0), \varepsilon_2 = (0, 1, 0), \varepsilon_3 = (0, 0, 1)$$

$$\theta = \{ \theta_1, \theta_2, \theta_3 \} \text{ in which } \theta_1 = (1, 1, 0), \theta_2 = (0, 1, 1), \theta_3 = (1, 0, 1)$$

- a) Find the transition matrix from the basis ε to the basis θ .
- b) Find the transition matrix from the basis θ to the basis ε .